

Math 526, F08
Quiz 9

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Are these pairs of vectors orthonormal or only orthogonal or only independent?

(a) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$

Independent only

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1 \neq 0$$

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 2 \neq 0$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1 \neq 0$$

(b) $\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}, \begin{bmatrix} 0.4 \\ 0.3 \end{bmatrix}.$

Orthogonal only

$$\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix} \cdot \begin{bmatrix} 0.4 \\ 0.3 \end{bmatrix} = 0$$

$$\text{but } \left\| \begin{bmatrix} -0.4 \\ 0.3 \end{bmatrix} \right\| = 0.5 \neq 1$$

(c) $\begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix}.$

orthonormal

$$\begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix} = \frac{1}{2} \cos \theta \sin \theta - \frac{1}{2} \sin \theta \cos \theta = 0$$

$$\left\| \begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix} \right\| = 1$$

$$\left\| \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix} \right\| = 1$$

Problem 2. (10 points) Find an orthonormal basis $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ for the column space of

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix} \text{ using the Gram-Schmidt process.}$$

$$\tilde{\mathbf{q}}_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \|\tilde{\mathbf{q}}_1\|^2 = 1$$

$$\tilde{\mathbf{q}}_2 = \mathbf{a}_2 - \left(\frac{\mathbf{a}_2^T \tilde{\mathbf{q}}_1}{\tilde{\mathbf{q}}_1^T \tilde{\mathbf{q}}_1} \right) \tilde{\mathbf{q}}_1, \quad \mathbf{a}_2^T \tilde{\mathbf{q}}_1 = 2$$

$$= \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \left(\frac{2}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}, \quad \|\tilde{\mathbf{q}}_2\| = 9$$

$$\tilde{\mathbf{q}}_3 = \mathbf{a}_3 - \left(\frac{\mathbf{a}_3^T \tilde{\mathbf{q}}_1}{\tilde{\mathbf{q}}_1^T \tilde{\mathbf{q}}_1} \right) \tilde{\mathbf{q}}_1 - \left(\frac{\mathbf{a}_3^T \tilde{\mathbf{q}}_2}{\tilde{\mathbf{q}}_2^T \tilde{\mathbf{q}}_2} \right) \tilde{\mathbf{q}}_2$$

$$= \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \left(\frac{4}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \left(\frac{18}{9} \right) \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

$$\mathbf{a}_3^T \tilde{\mathbf{q}}_1 = 4$$

$$\mathbf{a}_3^T \tilde{\mathbf{q}}_2 = 18$$

$$= \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$$

$$\Rightarrow \mathbf{q}_1 = \frac{\tilde{\mathbf{q}}_1}{\|\tilde{\mathbf{q}}_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{\tilde{\mathbf{q}}_2}{\|\tilde{\mathbf{q}}_2\|} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{q}_3 = \frac{\tilde{\mathbf{q}}_3}{\|\tilde{\mathbf{q}}_3\|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$